

CHAPTER

A

Hidden Markov Models

Chapter 8 introduced the Hidden Markov Model and applied it to part of speech tagging. Part of speech tagging is a fully-supervised learning task, because we have a corpus of words labeled with the correct part-of-speech tag. But many applications don't have labeled data. So in this chapter, we introduce the full set of algorithms for HMMs, including the key unsupervised learning algorithm for HMM, the Forward-Backward algorithm. We'll repeat some of the text from Chapter 8 for readers who want the whole story laid out in a single chapter.

A.1 Markov Chains

Markov chain

The HMM is based on augmenting the Markov chain. A **Markov chain** is a model that tells us something about the probabilities of sequences of random variables, *states*, each of which can take on values from some set. These sets can be words, or tags, or symbols representing anything, like the weather. A Markov chain makes a very strong assumption that if we want to predict the future in the sequence, all that matters is the current state. The states before the current state have no impact on the future except via the current state. It's as if to predict tomorrow's weather you could examine today's weather but you weren't allowed to look at yesterday's weather.

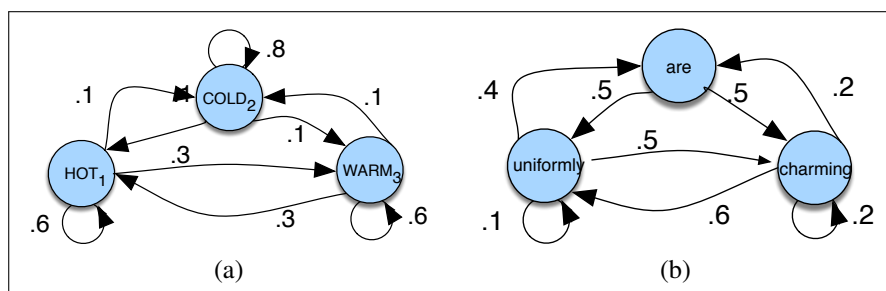


Figure A.1 A Markov chain for weather (a) and one for words (b), showing states and transitions. A start distribution π is required; setting $\pi = [0.1, 0.7, 0.2]$ for (a) would mean a probability 0.7 of starting in state 2 (cold), probability 0.1 of starting in state 1 (hot), etc.

Markov assumption

More formally, consider a sequence of state variables $q_1, q_2, \dots, q_i$. A Markov model embodies the **Markov assumption** on the probabilities of this sequence: that when predicting the future, the past doesn't matter, only the present.

$$\text{Markov Assumption: } P(q_i = a | q_1 \dots q_{i-1}) = P(q_i = a | q_{i-1}) \quad (\text{A.1})$$

Figure A.1a shows a Markov chain for assigning a probability to a sequence of weather events, for which the vocabulary consists of HOT, COLD, and WARM. The states are represented as nodes in the graph, and the transitions, with their probabilities, as edges. The transitions are probabilities: the values of arcs leaving a given

state must sum to 1. Figure A.1b shows a Markov chain for assigning a probability to a sequence of words $w_1 \dots w_n$. This Markov chain should be familiar; in fact, it represents a bigram language model, with each edge expressing the probability $p(w_i | w_j)$! Given the two models in Fig. A.1, we can assign a probability to any sequence from our vocabulary.

Formally, a Markov chain is specified by the following components:

$Q = q_1 q_2 \dots q_N$	a set of N states
$A = a_{11} a_{12} \dots a_{n1} \dots a_{nn}$	a transition probability matrix A , each a_{ij} representing the probability of moving from state i to state j , s.t. $\sum_{j=1}^n a_{ij} = 1 \quad \forall i$
$\pi = \pi_1, \pi_2, \dots, \pi_N$	an initial probability distribution over states. π_i is the probability that the Markov chain will start in state i . Some states j may have $\pi_j = 0$, meaning that they cannot be initial states. Also, $\sum_{i=1}^n \pi_i = 1$

Before you go on, use the sample probabilities in Fig. A.1a (with $\pi = [.1, .7, .2]$) to compute the probability of each of the following sequences:

(A.2) hot hot hot hot

(A.3) cold hot cold hot

What does the difference in these probabilities tell you about a real-world weather fact encoded in Fig. A.1a?

A.2 The Hidden Markov Model

A Markov chain is useful when we need to compute a probability for a sequence of observable events. In many cases, however, the events we are interested in are **hidden**: we don't observe them directly. For example we don't normally observe part-of-speech tags in a text. Rather, we see words, and must infer the tags from the word sequence. We call the tags **hidden** because they are not observed.

Hidden Markov model

A **hidden Markov model (HMM)** allows us to talk about both *observed* events (like words that we see in the input) and *hidden* events (like part-of-speech tags) that we think of as causal factors in our probabilistic model. An HMM is specified by the following components:

$Q = q_1 q_2 \dots q_N$	a set of N states
$A = a_{11} \dots a_{ij} \dots a_{NN}$	a transition probability matrix A , each a_{ij} representing the probability of moving from state i to state j , s.t. $\sum_{j=1}^N a_{ij} = 1 \quad \forall i$
$O = o_1 o_2 \dots o_T$	a sequence of T observations , each one drawn from a vocabulary $V = v_1, v_2, \dots, v_V$
$B = b_i(o_t)$	a sequence of observation likelihoods , also called emission probabilities , each expressing the probability of an observation o_t being generated from a state i
$\pi = \pi_1, \pi_2, \dots, \pi_N$	an initial probability distribution over states. π_i is the probability that the Markov chain will start in state i . Some states j may have $\pi_j = 0$, meaning that they cannot be initial states. Also, $\sum_{i=1}^n \pi_i = 1$

A first-order hidden Markov model instantiates two simplifying assumptions. First, as with a first-order Markov chain, the probability of a particular state depends only on the previous state:

$$\textbf{Markov Assumption: } P(q_i | q_1 \dots q_{i-1}) = P(q_i | q_{i-1}) \quad (\text{A.4})$$

Second, the probability of an output observation o_i depends only on the state that produced the observation q_i and not on any other states or any other observations:

$$\textbf{Output Independence: } P(o_i | q_1 \dots q_i, \dots, q_T, o_1, \dots, o_i, \dots, o_T) = P(o_i | q_i) \quad (\text{A.5})$$

To exemplify these models, we'll use a task invented by Jason [Eisner \(2002\)](#). Imagine that you are a climatologist in the year 2799 studying the history of global warming. You cannot find any records of the weather in Baltimore, Maryland, for the summer of 2020, but you do find Jason Eisner's diary, which lists how many ice creams Jason ate every day that summer. Our goal is to use these observations to estimate the temperature every day. We'll simplify this weather task by assuming there are only two kinds of days: cold (C) and hot (H). So the Eisner task is as follows:

Given a sequence of observations O (each an integer representing the number of ice creams eaten on a given day) find the 'hidden' sequence Q of weather states (H or C) which caused Jason to eat the ice cream.

Figure A.2 shows a sample HMM for the ice cream task. The two hidden states (H and C) correspond to hot and cold weather, and the observations (drawn from the alphabet $O = \{1, 2, 3\}$) correspond to the number of ice creams eaten by Jason on a given day.

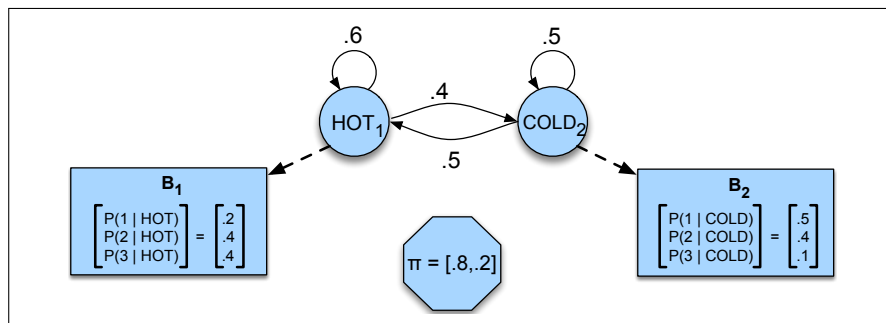


Figure A.2 A hidden Markov model for relating numbers of ice creams eaten by Jason (the observations) to the weather (H or C, the hidden variables).

An influential tutorial by [Rabiner \(1989\)](#), based on tutorials by Jack Ferguson in the 1960s, introduced the idea that hidden Markov models should be characterized by **three fundamental problems**:

Problem 1 (Likelihood):	Given an HMM $\lambda = (A, B)$ and an observation sequence O , determine the likelihood $P(O \lambda)$.
Problem 2 (Decoding):	Given an observation sequence O and an HMM $\lambda = (A, B)$, discover the best hidden state sequence Q .
Problem 3 (Learning):	Given an observation sequence O and the set of states in the HMM, learn the HMM parameters A and B .

We already saw an example of Problem 2 in Chapter 8. In the next two sections we introduce the Forward and Forward-Backward algorithms to solve Problems 1 and 3 and give more information on Problem 2